Source code :

Stack-

```
#include <iostream>
using namespace std;

class Stack {
private:
    int arr[100];
    int top;

public:
    Stack() {
        top = -1;
    }

    void insert(int value) { // Push
        if (top == 99) {
            cout << "Stack Overflow\n";
            return;
        }
        arr[++top] = value;
        cout << value << " inserted into stack.\n";
    }

    void deleteElement() { // Pop
        if (top == -1) {
            cout << "Stack Underflow\n";
            return;
        }
        cout << arr[top--] << " deleted from stack.\n";
    }

    void search(int value) {
        for (int i = 0; i <= top; i++) {
            if (arr[i] == value) {
                cout << value << " found in stack.\n";
                return;
            }
        }
        cout << value << " not found in stack.\n";
    }

    void display() {
        if (top == -1) {
            cout << "Stack is empty.\n";
```

```
            return;
        }

        cout << "Stack elements: ";
        for (int i = top; i >= 0; i--) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
};

int main() {
    Stack s;

    s.insert(10);
    s.insert(20);
    s.insert(30);

    s.display();

    s.search(20);

    s.deleteElement();

    s.display();

    return 0;
}
```

<u>Queue-</u>

```
#include <iostream>
using namespace std;

class Queue {
private:
    int arr[100];
    int front, rear;

public:
    Queue() {
        front = 0;
        rear = -1;
    }

    void insert(int value) { // Enqueue
        if (rear == 99) {
```

```cpp
            cout << "Queue Overflow\n";
            return;
        }

        arr[++rear] = value;
        cout << value << " inserted into queue.\n";
    }

    void deleteElement() { // Dequeue
        if (front > rear) {
            cout << "Queue Underflow\n";
            return;
        }

        cout << arr[front++] << " deleted from queue.\n";
    }

    void search(int value) {
        for (int i = front; i <= rear; i++) {
            if (arr[i] == value) {
                cout << value << " found in queue.\n";
                return;
            }
        }

        cout << value << " not found in queue.\n";
    }

    void display() {
        if (front > rear) {
            cout << "Queue is empty.\n";
            return;
        }

        cout << "Queue elements: ";
        for (int i = front; i <= rear; i++) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
};

int main() {
    Queue q;

    q.insert(10);
    q.insert(20);
    q.insert(30);
```

```
    q.display();

    q.search(20);

    q.deleteElement();

    q.display();

    return 0;
}
```